

Rajeet Ash

Kolkata, India • +91 9330515069 • rajeetash@hotmail.com • LinkedIn • Github • Portfolio

Robotics and Artificial Intelligence undergraduate with experience in full-stack systems, AI pipelines, and real-time architectures. Strong focus on scalable backend systems, optimization algorithms, and production-ready engineering.

TECHNICAL SKILLS

Programming: Kotlin, Python, JavaScript, TypeScript, C, C++, Java, SQL
AI/ML: LangChain (langchain-community), Google Gemini, Ollama, NLP Pipelines, cuDNN, OmniVerse
Web & Backend: Node.js, Express.js, Flask, REST APIs, WebSockets, BullMQ, Redis
Frontend: React, Next.js, Tailwind CSS, Vite
Database & Systems: PostgreSQL, SQLite, Neo4j (Graph DB), Supabase
Cloud & Tools: GCP, Azure Storage, Git, HuggingFace
Robotics: ROS2, Gemini Robotics-ER

EXPERIENCE

Web Development Lead Ureckon, UEM Kolkata	Mar 2026 – Present
Leading full-stack development and mentoring team members.	
Web Developer University of Engineering and Management	Dec 2025 – Mar 2026
Developed and maintained university web applications.	
Web Dev Lead & Co-Lead Organizer Technologia Hackathon, UEM Kolkata	Sep 2025 – Present
Led development and organization of large-scale hackathons.	
React Native Developer Intern Smile2Steps Research Project	May 2025 – Jun 2025
Built mobile components and integrated backend APIs.	

PROJECTS

JUKE - Android Music Streaming Application
- Built offline-first music streaming app with recommendation system. - Implemented MVVM architecture and local caching.
WAVES - Real-Time Tidal Prediction System
- Developed Android app for tidal prediction with Python data pipelines. - Designed backend-to-mobile data flow and caching.
BizChat - AI-Powered Business Communication Platform
- Built AI system to extract structured order data from chat and voice inputs. - Designed async architecture using Redis and BullMQ. - Integrated LLMs (Gemini, Ollama) for processing and summarization. - Implemented real-time updates and automated invoice generation.
Classroom Management System
- Built LMS with authentication and role-based access control. - Designed REST APIs and async processing.

RESEARCH

BE-PS Algorithm - Renewable Energy Data Processing
Designed entropy-based sorting achieving 2.67x speed improvement over C++.
PSO-Optimized Robotic Arm
- Solved inverse kinematics using Particle Swarm Optimization. - Achieved 10^{-12} simulation accuracy and hardware validation.

ACHIEVEMENTS

- Vice Chancellor's Award - Best Scientific Mind (2024–2025)
- Top 100 Finalist - National Coding League (17,000+ teams)

EDUCATION

B.Tech in Robotics and Artificial Intelligence

2024 – 2028

University of Engineering and Management, Kolkata

CGPA: 8.9/10

Class XII (ISC)

2023

Score: 90% | Computer Science: 98%

Class X (ICSE)

2021

Score: 92.7%